

Fig. P9.1 and P9.5

9.1, 5

$$y = \frac{h_2 - h_1}{a} x + h_1$$

$$\begin{aligned} I_y &= \int x^2 dA = \int_0^a x^2 y dx \\ &= \int_0^a \frac{h_2 - h_1}{a} x^3 + h_1 x^2 dx \\ &= \left. \frac{h_2 - h_1}{4a} x^4 + \frac{h_1}{3} x^3 \right|_0^a \\ &= \frac{h_2 - h_1}{4} a^3 + \frac{h_1}{3} a^3 \\ &= \left(\frac{h_2}{4} + \frac{h_1}{12} \right) a^3 \end{aligned}$$

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$$x = a \left(\frac{y - h_1}{h_2 - h_1} \right) = \frac{a}{h_2 - h_1} y - \frac{ah_1}{h_2 - h_1}$$

$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int y^2 x dy \\ &= \int_0^{h_2} \frac{h_2 - h_1}{h_2 - h_1} y^3 - \frac{ah_1}{h_2 - h_1} y^2 dy \\ &= \left. \frac{a}{4(h_2 - h_1)} y^4 - \frac{ah_1}{3(h_2 - h_1)} y^3 \right|_0^{h_2} \\ &= \frac{ah_2^4}{4(h_2 - h_1)} - \frac{ah_1 h_2^3}{3(h_2 - h_1)} \\ &= \frac{3ah_2^4 - 4ah_1 h_2^3}{12(h_2 - h_1)} \end{aligned}$$

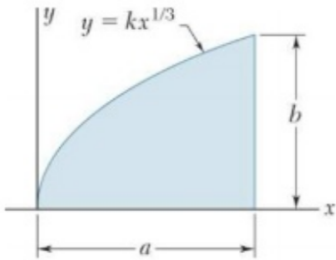


Fig. P9.2 and P9.6

4.2, 6

$$\begin{aligned} I_y &= \int x^2 dA \\ &= \int_0^a x^2 y dx \\ &= \int_0^a k x^{\frac{10}{3}} dx \\ &= \left. \frac{3}{10} k x^{\frac{16}{3}} \right|_0^a \\ &= \frac{3}{10} k a^{\frac{16}{3}} \end{aligned}$$

$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int y^2 x dy \\ &= \frac{1}{k^3} \int_0^b y^5 dy \\ &= \frac{1}{k^3} \times \frac{1}{6} y^6 \Big|_0^b \\ &= \frac{b^6}{6k^3} \end{aligned}$$

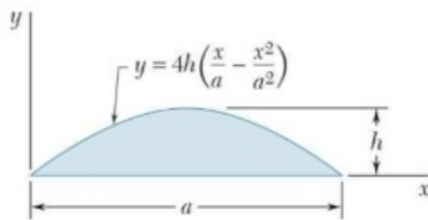


Fig. P9.3 and P9.7

9.3, 9.7

$$\begin{aligned} I_y &= \int x^2 dA \\ &= \int x^2 y dx \\ &= \int_0^a \frac{4h}{a} x^3 - \frac{4h}{a^2} x^4 dx \\ &= \left. \frac{h}{a} x^4 - \frac{4h}{5a^2} x^5 \right|_0^a \\ &= a^3 h - \frac{4}{5} a^3 h \\ &= \frac{a^3 h}{5} \end{aligned}$$

$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int_0^a y^3 dx \\ &= \int_0^a (4h)^3 \left(\frac{x}{a} \right)^3 \left(1 - \frac{x}{a} \right)^3 dx \\ &= \int_0^a (4h)^3 \frac{x^3}{a^3} \left(1 - 3\frac{x}{a} + 3\frac{x^2}{a^2} - \frac{x^3}{a^3} \right) dx \\ &= 64h^3 \int_0^a \left(\frac{x^3}{a^3} - 3\frac{x^4}{a^4} + 3\frac{x^5}{a^5} - \frac{x^6}{a^6} \right) dx \\ &= 64h^3 \left(\frac{1}{4} a - \frac{3}{5} a + \frac{3}{2} a - \frac{a}{7} \right) \\ &= 0.45ah^3 \end{aligned}$$

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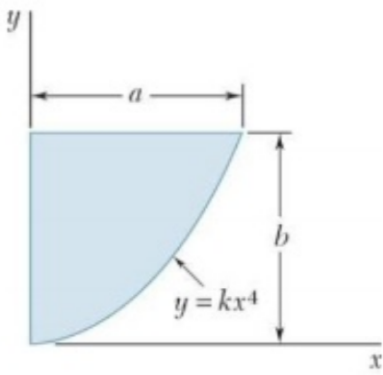


Fig. P9.4 and P9.8

$$\begin{aligned} 9.4, 8 \\ I_y &= \int x^2 dA \\ &= \int_0^a x^2 y dx \\ &= \int_0^a kx^6 dx \\ &= \frac{ka^7}{7} \end{aligned}$$

$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int_0^a y^3 dx \\ &= \int_0^a k^3 x^{12} dx \\ &= \frac{k^3 a^{13}}{13} \end{aligned}$$

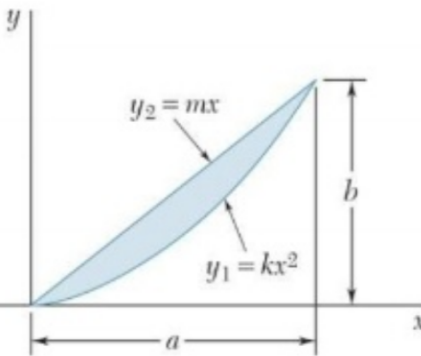
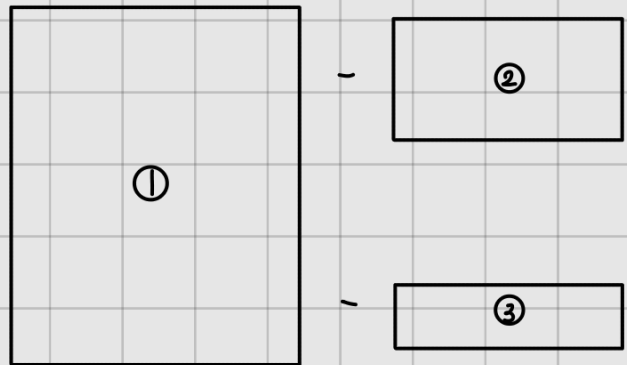
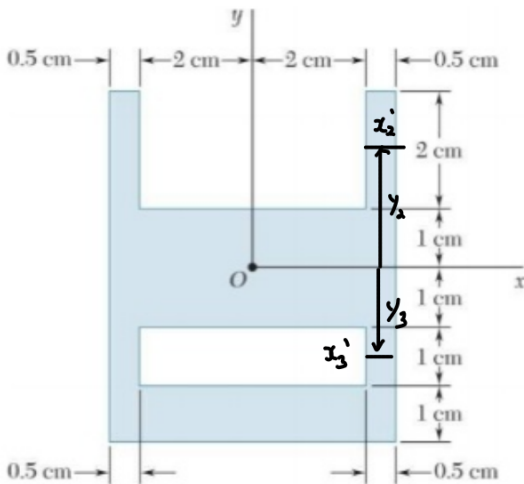


Fig. P9.9 and P9.12

$$\begin{aligned} 9.9, 12 \\ I_y &= \int x^2 dA \\ &= \int_0^a x^2 y dx \\ &= \int_0^a x^2 (mx - kx^2) dx \\ &= \left[\frac{m}{4} x^4 - \frac{k}{5} x^5 \right]_0^a \\ &= \frac{ma^4}{4} - \frac{ka^5}{5} \end{aligned}$$

$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int_0^a y^3 dx \\ &= \int_0^a (mx - kx^2)^3 dx \\ &= \int_0^a (m^3 x^3 - 3m^2 k x^4 + 3mk^2 x^5 - k^3 x^6) dx \\ &= \left[\frac{m^3}{4} x^4 - \frac{3m^2 k}{5} x^5 + \frac{1}{2} mk^2 x^6 - \frac{1}{7} k^3 x^7 \right]_0^a \\ &= \frac{1}{4} m^3 a^4 - \frac{3}{5} m^2 k a^5 + \frac{1}{2} mk^2 a^6 - \frac{1}{7} k^3 a^7 \end{aligned}$$

Problems 9.32, 9.34: Determine the moment of inertia and the radius of gyration of the shaded area wrt the x axis and the y axis.



$$I_{1x} = \frac{bh^3}{12} = \frac{5 \times 6^3}{12} = 90 \text{ cm}^4$$

$$I_{1y} = \frac{hb^3}{12} = \frac{6 \times 5^3}{12} = \frac{125}{2} \text{ cm}^4$$

$$I_{2x} = I_{2x'} + A_2 y_2^2 = \frac{bh^3}{12} + bh y_2^2 = \frac{4 \times 2^3}{12} + 4 \times 2 \times 2^2 = \frac{104}{3} \text{ cm}^4$$

$$I_{2y} = \frac{hb^3}{12} = \frac{2 \times 4^3}{12} = \frac{32}{3} \text{ cm}^4$$

$$I_{3x} = I_{3x'} + A_3 y_3^2 = \frac{bh^3}{12} + bh y_3^2 = \frac{4 \times 1^3}{12} + 4 \times 1 \times 1.5^2 = \frac{28}{3} \text{ cm}^4$$

$$I_{3y} = \frac{hb^3}{12} = \frac{1 \times 4^3}{12} = \frac{16}{3} \text{ cm}^4$$

$$I_x = I_{1x} - I_{2x} - I_{3x} =$$

$$I_y = I_{1y} - I_{2y} - I_{3y} =$$

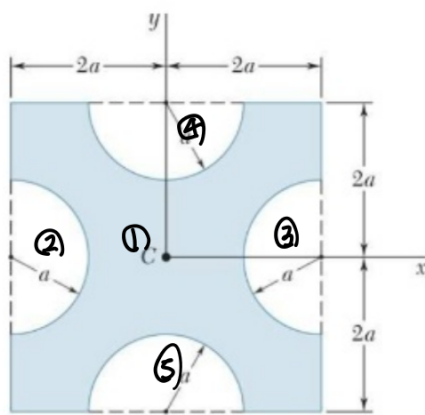
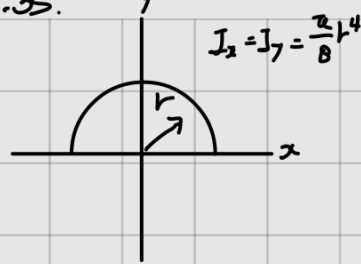


Fig. P9.35

9.35.



$$I_x = I_y = \frac{\pi}{8} r^4$$

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$$I_{1x} = \frac{bh^3}{12} = \frac{(4a)(4a)^3}{12} = \frac{64}{3} a^4$$

$$I_{1y} = \frac{hb^3}{12} = \frac{64}{3} a^4$$

$$I_{2x} = \frac{\pi}{8} a^4$$

$$I_{2y} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times (2a)^2 = \frac{17\pi}{8} a^4$$

$$I_{3x} = \frac{\pi}{8} a^4$$

$$I_{3y} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times (2a)^2 = \frac{17\pi}{8} a^4$$

$$I_{4x} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times (2a)^2 = \frac{17\pi}{8} a^4$$

$$I_{4y} = \frac{\pi}{8} a^4$$

$$I_{5x} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times (2a)^2 = \frac{17\pi}{8} a^4$$

$$I_{5y} = \frac{\pi}{8} a^4$$

$$\left(\begin{aligned} I_x &= I_{1x} - (I_{2x} + I_{3x} + I_{4x} + I_{5x}) = 7.146 a^4 \\ I_y &= I_{1y} - (I_{2y} + I_{3y} + I_{4y} + I_{5y}) = 7.146 a^4 \end{aligned} \right.$$

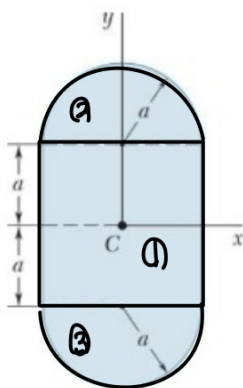


Fig. P9.36

9.36.

$$I_{1x} = \frac{bh^3}{12} = \frac{(2a)^3}{12} = \frac{4}{3} a^4$$

$$I_{1y} = \frac{hb^3}{12} = \frac{4}{3} a^4$$

$$I_{2x} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times a^2 = \frac{5\pi}{8} a^4$$

$$I_{2y} = \frac{\pi}{8} a^4$$

$$I_{3x} = \frac{\pi}{8} a^4 + \frac{\pi a^2}{2} \times a^2 = \frac{5\pi}{8} a^4$$

$$I_{3y} = \frac{\pi}{8} a^4$$

$$\left(\begin{aligned} I_x &= I_{1x} + I_{2x} + I_{3x} = 5.26 a^4 \\ I_y &= I_{1y} + I_{2y} + I_{3y} = 2.11 a^4 \end{aligned} \right.$$

Problems 9.45, 9.46: Determine the polar moment of inertia of the area shown wrt (a) point O, (b) the centroid of the area.

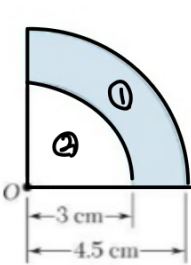


Fig. P9.45

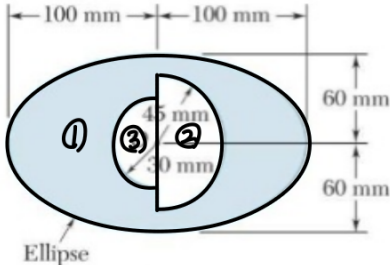


Fig. P9.46

9.45.

	$\bar{x}$	$\bar{y}$	A	$\bar{x}A$	$\bar{y}A$
①	$\frac{4}{3\pi} \times 4.5$	$\frac{4}{3\pi} \times 4.5$	$\frac{\pi}{4} \times 4.5^2$	30.375	30.375
- ②	$\frac{4}{3\pi} \times 3$	$\frac{4}{3\pi} \times 3$	$\frac{\pi}{4} \times 3^2$	9	9
Sum			0.635	21.375	21.375

$$\therefore \bar{x} = \frac{\sum \bar{x}_i A_i}{A} = 2.419 \text{ cm}$$

$$\bar{y} = \frac{\sum \bar{y}_i A_i}{A} = 2.419 \text{ cm}$$

Problems 9.45, 9.46: Determine the polar moment of inertia of the area shown wrt (a) point O, (b) the centroid of the area.

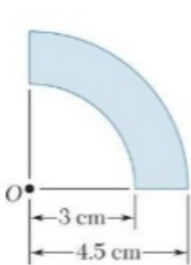


Fig. P9.45

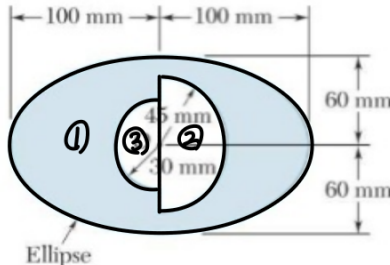


Fig. P9.46

	$\bar{x}$	$\bar{y}$	A	$\bar{x}A$	$\bar{y}A$
①	0	0	$\pi \times 100 \times 60$	0	0
- ②	$\frac{4}{3\pi} \times 45$	0	$\frac{\pi}{2} \times 45^2$	66750	0
- ③	$-\frac{4}{3\pi} \times 45$	0	$\frac{\pi}{2} \times 30^2$	-27000	0
Sum			$\frac{9075}{2} \pi$	-33750	0

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{A} = -2.367 \text{ mm}$$

$$\bar{y} = 0$$

$$I_{1x} = \frac{\pi}{16} (4.5)^4$$

$$I_{1y} = \frac{\pi}{16} (4.5)^4$$

$$I_{2x} = \frac{\pi}{16} \times 3^4$$

$$I_{2y} = \frac{\pi}{16} \times 3^4$$

$$I_x = I_{1x} - I_{2x} = 64.61 \text{ cm}^4$$

$$I_y = I_{1y} - I_{2y} = 64.61 \text{ cm}^4$$

$$I_{\bar{x}} = I_x - A \bar{y}^2 = 12.91 \text{ cm}^4$$

$$I_{\bar{y}} = I_y - A \bar{x}^2 = 12.91 \text{ cm}^4$$

$$I_{1x} = \frac{1}{4} \times \pi \times 100 \times 60^3 = 16964600$$

$$I_{1y} = \frac{1}{4} \times \pi \times 100^3 \times 60 = 47123889.8$$

$$I_{2x} = \frac{1}{8} \times \pi \times 30^4 = 318086$$

$$I_{2y} = 318086$$

$$I_{3x} = \frac{1}{8} \times \pi \times 45^4 = 1610311$$

$$I_{3y} = 1610311$$

$$a) \begin{cases} I_x = I_{1x} - I_{2x} - I_{3x} = 15036203 \text{ mm}^4 \\ I_y = I_{1y} - I_{2y} - I_{3y} = 45195492 \text{ mm}^4 \end{cases}$$

$$b) \begin{cases} I_{\bar{x}} = I_x \pm A \bar{y}^2 = I_x = 15036203 \text{ mm}^4 \\ I_{\bar{y}} = I_y \pm A \bar{x}^2 = I_y + \frac{9075}{2} \pi \times (2.367)^2 \\ = 45275658.2 \text{ mm}^4 \end{cases}$$

Problems 9.111: A thin plate of mass m is cut in the shape of an equilateral triangle of side a . Determine the mass moment of inertia of the plate wrt (a) the centroidal axes AA' and BB' , (b) the centroidal axis CC' that is perpendicular to the plate.

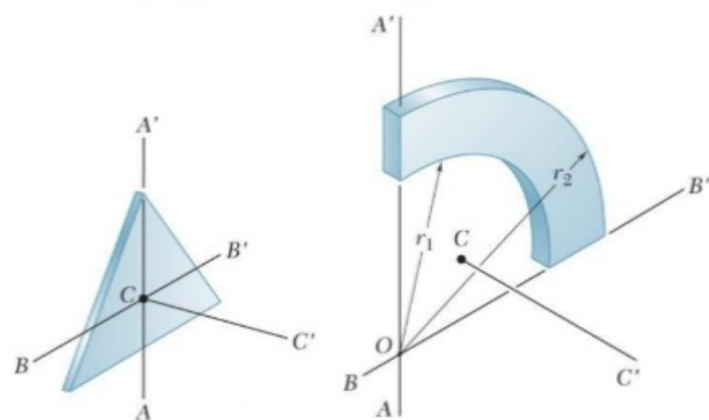


Fig. P9.111

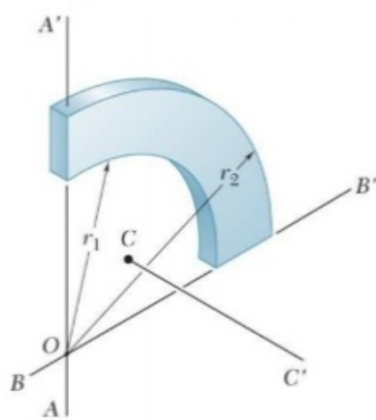
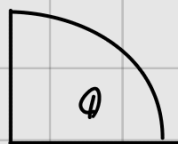


Fig. P9.114

Problems 9.114: The quarter ring shown has a mass m and was cut from a thin, uniform plate. Knowing that $r_1 = \frac{3}{4}r_2$, determine the mass moment of inertia of the quarter ring wrt (a) the axis AA' , (b) the centroidal axis CC' that is perpendicular to the plane of the quarter ring.

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$$I_{AA',1} = \frac{\pi}{4} r_2^4$$

$$A_1 = \frac{\pi}{4} r_2^2$$

$$I_{AA',2} = \frac{\pi}{16} r_1^4$$

$$A_2 = \frac{\pi}{4} r_1^2$$

$$I_{AA'} = I_{AA',1} - I_{AA',2} = \frac{\pi}{16} (r_2^4 - r_1^4)$$

$$A = A_1 - A_2 = \frac{\pi}{4} (r_2^2 - r_1^2)$$

$$I_{BB'} = I_{AA'} = \frac{\pi}{16} (r_2^4 - r_1^4)$$

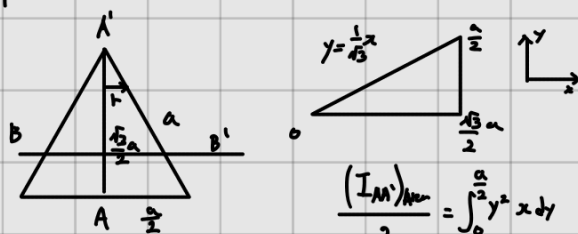
$$I_{AA',mass} = \frac{m}{A} I_{AA'} = m \times \frac{4}{\pi} \times \frac{1}{(r_2^2 - r_1^2)} \times \frac{\pi}{16} (r_2^4 - r_1^4)$$

$$= \frac{m}{4} (r_2^2 + r_1^2)$$

$$I_{BB',mass} = \frac{m}{A} I_{BB'} = I_{AA',mass} = \frac{m}{4} (r_2^2 + r_1^2)$$

$$I_{CC',mass} = I_{AA',mass} + I_{BB',mass} = \frac{m}{2} (r_2^2 + r_1^2)$$

9.111



$$A = a \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{4} a^2$$

$$\frac{(I_{AA'})_{Area}}{2} = \int_0^{\frac{a}{2}} y^2 \times dy$$

$$= \int_0^{\frac{a}{2}} \sqrt{3} y^2 dy$$

$$= \frac{\sqrt{3}}{4} y^3 \Big|_0^{\frac{a}{2}}$$

$$(I_{AA'})_{Area} = \frac{\sqrt{3}}{32} a^4$$

$$(I_{BB'})_{Area} = \frac{1}{36} \times a \times \left(\frac{\sqrt{3}a}{2}\right)^3 = \frac{\sqrt{3}}{96} a^4$$

$$(I_{AA'})_{mass} = \int r^2 dm = \int r^2 (I_{AA'})_{Area}$$

$$= \frac{m}{A} \times \frac{\sqrt{3}}{32} a^4 = ma^4 \times \frac{4}{\sqrt{3}a^2} \times \frac{\sqrt{3}}{32}$$

$$= \frac{1}{8} ma^2$$

$$(I_{BB'})_{mass} = \frac{m}{A} (I_{BB'})_{Area}$$

$$= m \times \frac{4}{\sqrt{3}a^2} \times \frac{\sqrt{3}}{96} a^4$$

$$= \frac{1}{24} ma^2$$

$$(I_{CC'})_{mass} = \frac{m}{A} (I_{CC'})_{Area}$$

$$= \frac{m}{A} (I_{AA',Area} + I_{BB',Area})$$

$$= m \times \frac{4}{\sqrt{3}a^2} \times \left(\frac{\sqrt{3}}{32} a^4 + \frac{\sqrt{3}}{96} a^4 \right)$$

$$= \frac{1}{6} ma^2$$

Problems 9.119: Determine by direct integration the mass moment of inertia wrt the z axis of the right circular cylinder shown, assuming that it has a uniform density and a mass m .

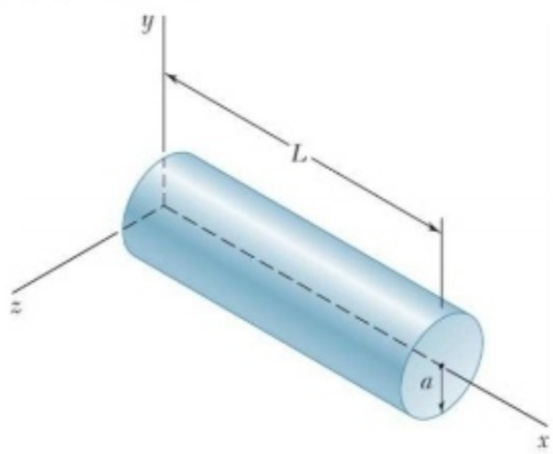
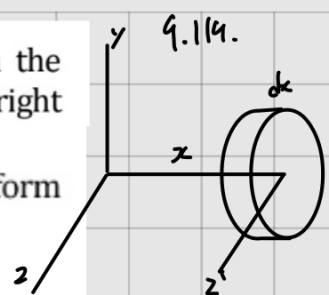


Fig. P9.119



9.119.

$$dm = \rho \omega^2 dx$$

$$dI_{z'} = \frac{a^2}{4} dm$$

$$I_{z'} = dI_{z'} + x^2 dm$$

$$dI_z = \rho \left(\frac{a^2}{4} \omega^2 dx + x^2 \omega^2 dx \right)$$

$$I_z = \rho \int_0^L \left(\frac{\omega a^4}{4} + \omega x^2 \right) dx$$

$$= \rho \omega \left(\frac{a^4 L}{4} + \frac{a^2 L^3}{3} \right)$$

Problems 9.122, 9.123: Determine by direct integration the mass moment of inertia wrt the x axis and the y axis of the pyramid shown, assuming that it has a uniform density and a mass m .

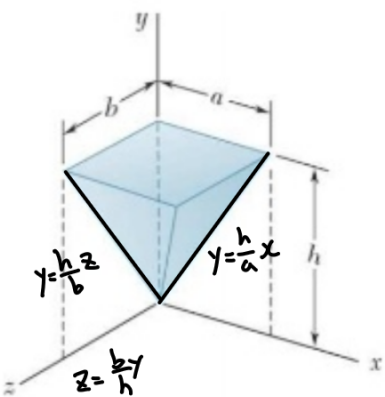


Fig. P9.122 and P9.123

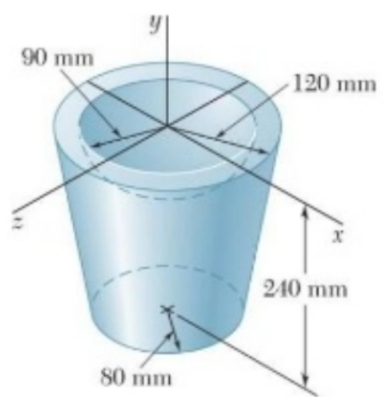


Fig. P9.131

$$dm = \rho A dy$$

$$= \rho x z dy$$

$$= \rho \times \frac{a}{h} y \times \frac{b}{h} y dy$$

$$= \rho \frac{ab}{h^2} y^2 dy$$

$$dI_{x'} = z^2 dm$$

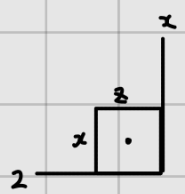
$$dI_x = dI_{x'} + y^2 dm$$

$$I_x = \int z^2 dm + \int \rho \frac{ab}{h^2} y^4 dy$$

$$= \int_0^h \frac{b^2}{h^2} y^2 \times \rho \frac{ab}{h^2} y^2 dy + \int_0^h \rho \frac{ab}{h^2} y^4 dy$$

$$= \int_0^h \rho \frac{ab^3}{h^4} y^4 dy + \int_0^h \rho \frac{ab}{h^2} y^4 dy$$

$$= \left(\rho \frac{ab^3}{h^4} + \rho \frac{ab}{h^2} \right) \frac{h^5}{5}$$



Problems 9.131: The steel machine component shown is formed by machining a hemisphere into the base of a truncated cone. Knowing that the density of steel is 7850 kg/m^3 , determine the mass moment of inertia of the component wrt the y axis.

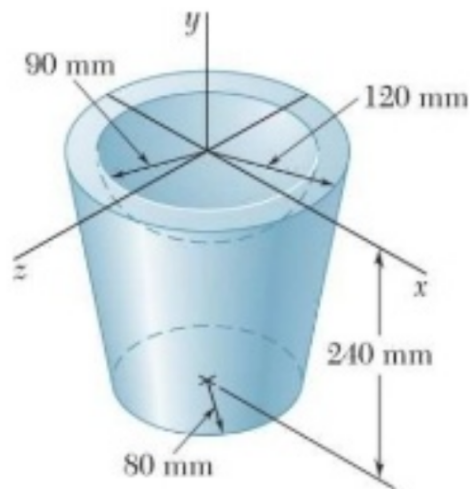
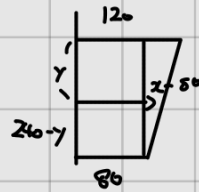


Fig. P9.131



$$\frac{240-y}{x-80} = \frac{240}{120-80}$$

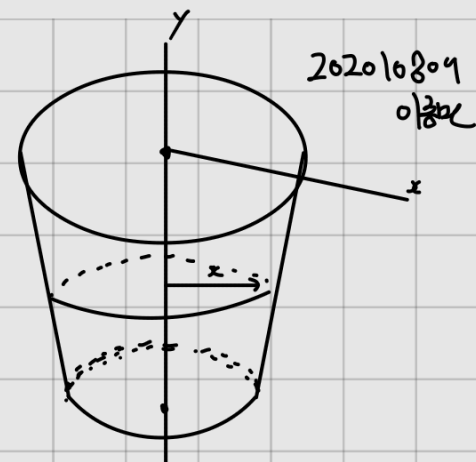
$$240-y = 6x-480$$

$$(m) \quad 0.12 = 6x+y$$

$$0 = 6\frac{dx}{dy} + 1$$

$$dy = -6dx$$

truncated cone



$$dI_y = \frac{1}{2} x^2 dm$$

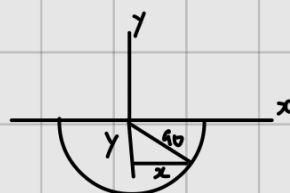
$$dm = \rho \pi x^2 dy$$

$$I_{y, \text{cone}} = \int dI_y = \int_0^{0.24} \frac{1}{2} x^2 \times \rho \pi x^2 dy$$

$$= \int_{0.08}^{0.12} 3\rho \pi x^4 dx$$

$$= 0.3197$$

Hemisphere



$$x = \sqrt{0.09^2 - y^2}$$

$$x^4 = (0.09^2 - y^2)^2$$

$$dI_y = \frac{1}{2} x^2 dm$$

$$dm = \rho \pi x^2 dy$$

$$I_y = \int dI_y = \int_0^{0.09} \frac{1}{2} x^2 \times \rho \pi x^2 dy$$

$$= \frac{\rho \pi}{2} \int_0^{0.09} (0.09^2 - y^2)^2 dy$$

$$= 0.0388$$

$$\therefore I_y = I_{y, \text{cone}} - I_{y, \text{hemisphere}} = 0.2808$$

Problems 9.138: A .0025 m-thick piece of sheet steel is cut and bent into the machine component shown. Knowing that the density of steel is 7850 kg/m^3 , determine the mass moment of inertia of the component wrt each of the coordinate axes.

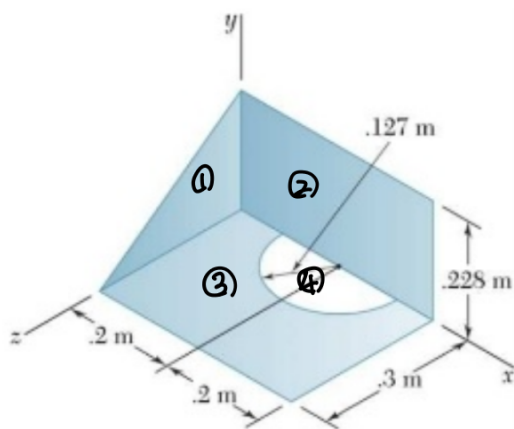
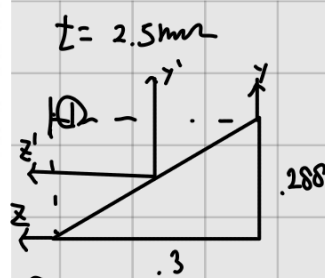


Fig. P9.138



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$$① \quad I_x = \frac{m_1}{24} (0.3^2 + 0.228^2) + m_1 (0.15^2 + 0.144^2)$$

$$I_y = \frac{m_1}{24} (0.3)^2 + m_1 \times (0.15)^2$$

$$I_z = \frac{m_1}{24} (0.228^2) + m_1 (0.144^2)$$

$$② \quad I_x = \frac{1}{12} m_2 (0.228)^2 + m_2 \left(\frac{0.228}{2} \right)^2$$

$$I_y = \frac{1}{12} m_2 (0.4)^2 + m_2 (0.2)^2$$

$$I_z = \frac{1}{12} m_2 (0.228^2 + 0.4^2) + m_2 \times (0.2^2 + 0.114^2)$$

$$③ \quad I_x = \frac{1}{12} m_3 (0.3)^2 + m_3 \times (0.15^2)$$

$$I_y = \frac{1}{12} m_3 (0.3^2 + 0.4^2) + m_3 \times (0.15^2 + 0.2^2)$$

$$I_z = \frac{1}{12} m_3 \times (0.4^2) + m_3 \times (0.2)^2$$

$$④ \quad I_x = \frac{1}{8} m_4 (0.127)^2$$

$$I_y = \frac{1}{4} m_4 (0.127)^2 + m_4 (0.2)^2$$

$$I_z = \frac{1}{8} m_4 (0.127)^2 + m_4 \times (0.2)^2$$

$$\rho = 7850 \times 0.0025$$

$$= \frac{157}{8}$$

$$m_1 = \rho A_1 = \rho \times (0.3) (0.228) \times \frac{1}{2} = 0.671175$$

$$m_2 = \rho A_2 = \rho \times (0.228) (0.127) = 0.5682615$$

$$m_3 = \rho A_3 = \rho \times (0.4) (0.3) = 2.355$$

$$m_4 = \rho A_4 = \rho \times \pi \times (0.127)^2 \times \frac{1}{2} = 0.497206$$

$$\left\{ \begin{array}{l} I_x = I_{x,1} + I_{x,2} + I_{x,3} - I_{x,4} = 0.1 \text{ kg} \cdot \text{m}^2 \\ I_y = I_{y,1} + I_{y,2} + I_{y,3} - I_{y,4} = 0.222 \text{ kg} \cdot \text{m}^2 \\ I_z = I_{z,1} + I_{z,2} + I_{z,3} - I_{z,4} = 0.153 \text{ kg} \cdot \text{m}^2 \end{array} \right.$$

Problems 9.141: The machine element shown is fabricated from steel. Knowing that the density of steel is 7850 kg/m^3 , determine the mass moment of inertia of the assembly wrt each of the coordinate axes.

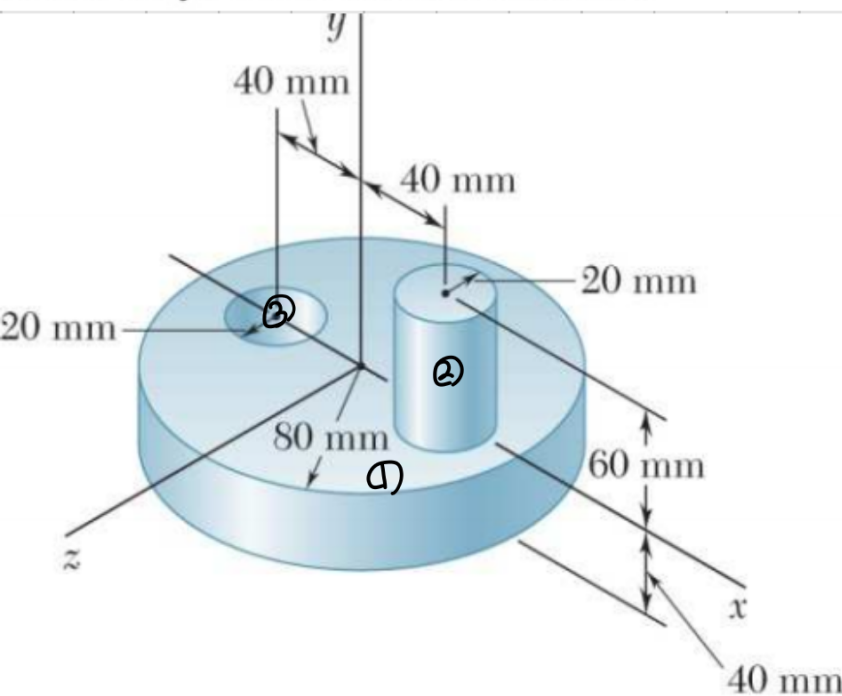


Fig. P9.141

$$\textcircled{1} \quad I_x = \frac{1}{12} m_1 (3 \times 80^2 + 40^2) + m_1 \times 20^2 \quad \boxed{\text{mm} \rightarrow \text{m}}$$

$$I_y = \frac{1}{12} m_1 \times 80^2$$

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$$I_z = \frac{1}{12} m_1 (3 \times 80^2 + 40^2) + m_1 \times 20^2$$

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$$\textcircled{2} \quad I_x = \frac{1}{12} m_2 (3 \times 20^2 + 60^2) + m_2 \times 30^2$$

$$I_y = \frac{1}{12} m_2 \times 20^2 + m_2 \times 40^2$$

$$I_z = \frac{1}{12} m_2 (3 \times 20^2 + 60^2) + m_2 \times 50^2$$

$$\textcircled{3} \quad I_x = \frac{1}{12} m_3 (3 \times 20^2 + 40^2) + m_3 \times 20^2$$

$$I_y = \frac{1}{12} m_3 \times 20^2 + m_3 \times 40^2$$

$$I_z = \frac{1}{12} m_3 (3 \times 20^2 + 40^2) + m_3 \times 44.72^2$$

$$m_1 = \rho \pi \times 80^2 \times 60 = 6.31 \text{ kg}$$

$$m_2 = \rho \pi \times 20^2 \times 60 = 0.591 \text{ kg}$$

$$m_3 = \rho \pi \times 20^2 \times 40 = 0.394 \text{ kg}$$

$$I_x = I_{x,1} + I_{x,2} - I_{x,3} = 0.01398 \text{ kg} \cdot \text{m}^2$$

$$I_y = I_{y,1} + I_{y,2} - I_{y,3} = 3.59 \times 10^{-3} \text{ kg} \cdot \text{m}^2$$

$$I_z = I_{z,1} + I_{z,2} - I_{z,3} = 0.61191 \text{ kg} \cdot \text{m}^2$$